

Trainer Debrief — Lab 22

Solution Generation, Selection and Lean Countermeasures — In Depth

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Generate and select improvements targeting your proven root causes (A5).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-22-solution-selection.xlsx

Two candidates must be struck immediately: 'Shop floor air conditioning' attacks a cause Lab 21 DISPROVED, and 'Replace all six welders' is a SGD 1.45m capital solution that no evidence supports. '100% burst testing' is the seductive wrong answer — it is pure inspection, it adds SGD 240k of cost, it does not reduce variation, and it is the opposite of what Control means in Six Sigma. The winning set is error-proofing first: clamp pressure interlock, PLC recipe lock and stroke counter — together SGD 99,000, they attack all three proven Xs, and they hold without anyone remembering to do anything. Note the hierarchy: ERROR-PROOF beats CONTROL beats TRAINING, because training decays the moment the trainer leaves the room.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Every solution traces to a proven root cause, criteria weights are agreed with the sponsor, and your selected set collectively addresses the majority of the quantified Y gap.

Where this sits in the capstone

Re-derive solution generation and the full Lean countermeasure toolkit, then select rigorously with a weighted matrix. Every solution must attack a PROVEN root cause from Lab 21 — solutions that address unproven causes are how projects quietly fail. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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